

Joana Mizrahi

joana.mizrahi@gmail.com • [linkedin.com/in/joana-mizrahi-336620300](https://www.linkedin.com/in/joana-mizrahi-336620300) • joanamizrahi.github.io

Robotics, Controls & Autonomy | Stanford M.S. Electrical Engineering, graduating June 2027

EDUCATION

Stanford University, Stanford, California June 2027

M.S. in Electrical Engineering, depth in Control and Optimization

GPA 4.10

- Coursework: Experimental Robotics; Optimal & Learning-Based Control; Robot Perception; State Estimation & Filtering; Convex Optimization; Computational Imaging; Decision Making under Uncertainty; Deep Learning
- Projects: violin-playing robot on a Flexiv Rizon arm (CS 225A); vision-based vehicle tracking on KITTI with a particle filter and multi-cue likelihoods (AA 273); grasp-success prediction from 3D point clouds with PointNet (CS 230)

Carnegie Mellon University, Pittsburgh, Pennsylvania

May 2025

Exchange Program, Electrical and Computer Engineering

GPA 4.0

- One of 4 students selected for a 1-year exchange; grant from EPFL and Severin B. Hacker (Duolingo co-founder and CTO)
- Projects: embedded RTOS on STM32 (interrupts, scheduling, motor control); MEMS dual-axis micromirror; neural amplifier

École Polytechnique Fédérale de Lausanne (EPFL), Lausanne, Switzerland

May 2025

B.Sc. in Microengineering

GPA 5.74/6

- Valedictorian: awarded the EPFL “Wafer d’Or” for the highest GPA in the Microengineering Bachelor’s program
- SHS Laureate: best Global Issues poster, 2023
- Projects: C++ robot simulation; embedded systems in AVR assembly; flexure-based marine chronometer oscillator

EXPERIENCE

Stanford Robotics Center, Stanford, California

June 2026 – September 2026

Robotics Intern

- Developed WorldNav, a framework for training outdoor navigation policies with on-policy RL inside a generative world model built from a single walkthrough video, for the Unitree Go2W quadruped
- Evaluation: extended the world model to generate terrain semantics together with each image, giving the policy a traversability reward that stays consistent with what it sees
- Exploration: bounded where the policy may train using measures of how well each generated view is supported by the recording, so it cannot exploit unreliable generations

Stanford Robotics Center, Stanford, California

April 2026 – present

Graduate Researcher. Mentor: Jing Liang. Research units with Prof. Jiajun Wu (CS 399) and Prof. Marco Pavone (AA 290)

- Co-first author of the WorldNav paper (in preparation for CVPR 2027); built the initial pipeline that turns monocular video into a semantic 3D simulator, now evaluating the world model and policies on campus scenes
- Voice navigation on the Unitree Go2W: designed and implemented a real-time behavior-tree task planner with an OpenAI Realtime API voice interface and open-vocabulary hazard alerting (ROS 2)

Virtual Traffic Lights, Pittsburgh, Pennsylvania

August 2024 – January 2026

Software Engineer

- Integrated traffic signal controller hardware (Econolite ASC/3, NTCIP over SNMP) with the company’s ML signal-timing optimizer for real-time closed-loop testing
- Built a backend service to version, schedule and approve AI-generated timing plans

TEACHING

Course Assistant, AA228/CS238 Decision Making under Uncertainty, Stanford

September 2026 – present

Grader, EE 102A Signals and Systems I, Stanford

January 2026 – March 2026

Teaching Assistant, PHYS-101 Physics and MSE-101 Materials Science, EPFL

September 2023 – June 2024

SKILLS

Programming: Python, C/C++, MATLAB

Robotics & ML: ROS 2, PyTorch, reinforcement learning, video diffusion fine-tuning, Gaussian splatting, state estimation

Software & hardware: Git, Docker, gRPC; CATIA, Fusion 360, COMSOL, LTspice; LaTeX

Languages: French (native), English (fluent), German (basic)

ACTIVITIES

Competitive badminton (national tournaments, CMU team); mathematics and French tutor (ARA)